13. *See id.* This company reported that the combined cost for producing their first TCFD, SASB, and GRI disclosures was between $200,000 and $350,000 but did not provide the cost for TCFD only. However, it noted that the cost of preparing its first CDP questionnaire did not exceed $50,000. To be conservative, we assume the initial CDP-related cost is $50,000. We apply the 4-to-9 ratio to convert this to the initial costs of TCFD disclosure and then apply a 44% reduction to estimate the ongoing cost. $50,000*(9/4)*(1 – 0.44) = $63,000.

14. *See* letter from AEPC. This commenter stated that initial costs to report in alignment with certain aspects of the TCFD can be around $500,000. To estimate ongoing costs, we refer to the percentage reduction from initial to ongoing costs as reflected by the FCA rule (56%) and the BEIS rule (32%), of which the median is 44%. $500,000*(1 – 0.44) = $280,000.

15. *See supra* note 3021. This source, which provided indicative fees for TCFD reporting services, noted that the average cost would be around $100,000. To estimate ongoing costs, we refer to the percentage reduction from initial to ongoing costs as reflected by the FCA rule (56%) and the BEIS rule (32%), of which the median is 44%. $100,000*(1 – 0.44) = $56,000.

16. When there is an even number of data points, there is no single middle value. In such cases, the median is computed as the arithmetic mean of the two middle data points. Accordingly, the median of Column (6) is calculated as follows: ($86,270 + $280,000)/2 = $183,135.

We next discuss our estimation process and methodology involved in producing the numbers in Table 8, including which cost estimates were included versus excluded, what assumptions were incorporated, and how the adjustment factors for targets and goals and GHG emissions measurement were calculated and applied. Many commenters did not explicitly state whether the costs of measuring emissions or setting targets and goals were included in their TCFD costs. As a result, we assumed that such costs were included only if such activities were contained in their qualitative description of climate-related disclosure activities. Of the twelve cost estimates presented in Table 8, we assume that three included the cost of target-related

activities, as indicated in Column (4).[3050] We also assume that five estimates included the costs of measuring Scope 1, Scope 2, and some Scope 3 emissions, as indicated in Column (5).[3051]

We next review cost estimates specific to setting targets and goals and assessing GHG emissions (Scope 1, Scope 2, and some Scope 3) from other sources in order to obtain their adjustment factors ($54,015 and $79,236, respectively). We recognize that the final rules do not necessarily require registrants to incur costs associated with setting targets and goals or measuring all three scopes of GHG emissions. We review such cost estimates because we determined that some of the sources in Table 8 included them with their overall TCFD-related costs; however, they should not necessarily be interpreted as direct compliance costs resulting from the final rules. Instead, we use these cost estimates to obtain appropriate adjustment factors that are subsequently subtracted from the applicable estimates in Column (3).

3050 The Financial Services company stated that it "reports on its progress towards its low-carbon financing and carbon-neutrality goals; the percentage of renewable energy sourced to support its operations and the percentage of energy reductions year over year." The Communication Services company reports that it gathers metrics/data related to carbon abatement, renewable energy, water conservation, and incurs expenses for monitoring and data quality. (*See* letter from Soc. Corp. Gov (June 11, 2021)). Company 6 stated that it "gathers data and reports on progress towards the company's low-carbon financing goal, progress toward the company's carbon-neutrality goal, the percentage of renewable energy sourced to support the company's operations, the percentage of energy reduction year-over-year." (*See* letter from Soc. Corp. Gov (June 17, 2022)). Based on the description of these activities, we assume that these three companies included the costs of setting targets and goals in their reported costs. The Large-cap financial institution stated that it "is committed to achieving net-zero emissions by 2050 and is in the process of implementing the Paris Agreement Capital Transition Assessment (PACTA) methodology to align its loan portfolio with the goals of the Paris Agreement." *See supra* note 2987. However, given their relatively low reported costs, we assume that the cost of setting targets and goals is not included in order to remain conservative in our estimation.

3051 The Financial Services company, Communications Services company, Company 5, and Company 6 all explicitly state that they measure and report Scopes 1, 2, and some Scope 3 emissions. The Energy company does not explicitly state that it measures emissions, however it states that it requires "consultants in emissions, climate science, and modeling," "multiple engineering disciplines," and "GHG emissions reporting expertise" as part of its disclosures. *See* letters from Soc. Corp. Gov (June 11, 2021 and June 17, 2022). Thus, we assume that these five companies included the costs of measuring Scope 1, 2, and some Scope 3 emissions within their reported costs. The Large-cap financial institution and Mid-cap company also report measuring the three scopes of emissions, however given their relatively low reported costs, we assume that the cost of measuring emissions is not included in order to remain conservative in our estimation.

The adjustment factor for setting targets and goals is $54,015, as indicated in Column (4). To obtain this number, we begin by reviewing four sources that provided more specific cost estimates related to targets and goals,[3052] which are presented in Table 9. The BEIS rule estimated that Metrics and Targets (including the cost of data gathering and cost of reporting, unrelated to GHG emissions) would have an ongoing cost of $72,359, while the FCA rule estimated the ongoing cost to be $53,507.

The remaining two sources only provided initial costs. Thus, to estimate the ongoing cost, we referred to the percent reduction from initial to ongoing costs reflected by the BEIS rule (23 percent reduction) and FCA rule (67 percent reduction), which yields a median percent reduction of 45 percent. One source estimated that setting targets would come with an initial cost ranging from $20,000 to $30,000.[3053] We apply the 45 percent reduction to arrive at an ongoing cost estimate of $13,750. Another company reported that it spent $1 million as an initial cost for target baseline and projections.[3054] We similarly apply the 45 percent reduction to arrive at an ongoing cost estimate of $550,000. The median of the ongoing costs of setting targets in Table 9 is $54,015, which is used as the adjustment factor for setting targets and goals (as indicated in Column (4) of Table 9).

[3052] These sources generally do not provide sufficient detail on precisely what the targets and goals disclosure would consist of; therefore, it is difficult to determine to what extent the corresponding cost estimates are applicable to the final rules' requirements on targets and goals. We can nevertheless use these sources to help us arrive at better informed compliance cost estimates. Similar reasoning can be applied to the cost estimates of scenario analysis, discussed in section IV.C.3.b.iv.

[3053] *See supra* note 3021.

[3054] *See* letter from Soc. Corp. Gov. (June 17, 2022).

Table 9. Ongoing costs of setting targets and goals

Commenter or Source	Ongoing costs of setting targets and goals
BEIS rule	$72,359 [1]
FCA rule	$35,671 [2]
Society for Corporate Governance (June 17, 2022)	$550,000 [3]
S&P Global	$13,750 [4]

1. *See supra* note 2996. The BEIS rule estimated that the ongoing cost metrics and targets disclosure is £58,500 (£52,000 for annual data gathering and £6,500 for the cost of reporting). We apply the 2022 average exchange rate. (£58,500)*(1.2369 USD/GBP) = $72,359. This reflects a 23% reduction from initial to ongoing costs, which we consider in determining the appropriate percentage reduction in subsequent calculations.
2. *See supra* note 2995. The FCA rule estimated that ongoing costs for metrics and targets disclosure is £43,259; however, this figure includes assessing Scopes 1 and 2 emissions. The corresponding initial cost disaggregates the cost, with two-thirds allocated to metrics and targets unrelated to Scopes 1 and 2 emissions. We assume the ongoing cost reflects the same proportional allocation and then we apply the 2022 average exchange rate. (£43,259)*(2/3)*(1.2369 USD/GBP) = $35,671. This reflects a 67% reduction from initial to ongoing costs, which we consider in determining the appropriate percentage reduction in subsequent calculations.
3. *See* letter from Soc. Corp. Gov (June 17, 2022). "Company 7" in this comment letter reported that it spent $1 million on "building a database for target baseline and projections," but did not provide the ongoing cost. To estimate the ongoing cost, we refer to the percentage reduction from initial to ongoing costs as reflected by the BEIS rule (23%) and the FCA rule (67%), of which the median is 45%. ($1,000,000)*(1 – 0.45) = $550,000.
4. *See supra* note 3021. The S&P Global meeting memorandum provides estimates on the initial cost of setting target ($20,000 - $30,000) but does not provide estimates with respect to the ongoing cost. To estimate the ongoing cost, we refer to the percentage reduction from initial to ongoing costs as reflected by the BEIS rule (23%) and the FCA rule (67%), of which the median is 45%. We apply this median percentage reduction to the midpoint of the initial cost: ($25,000)*(1 – 0.45) = $13,750.

Next, we focus on the adjustment factor for assessing Scope 1, Scope 2, and some Scope 3 emissions (as indicated in Column (5) of Table 8). To obtain this number, we review eight relevant estimates, which are presented in Table 10. Where necessary, modifications or assumptions are applied to the estimates (see table footnotes for details). Lastly, we take the median of these eight data points to obtain the adjustment factor for measuring Scope 1, Scope 2, and some Scope 3 emissions: $79,236. We reiterate that the final rules do not require the disclosure of Scope 1 and 2 emissions in all cases or from all registrants, and Scope 3 disclosures are not required. We reviewed these emissions cost in this section because we subtract them

from applicable estimates in Column (3) of Table 8, which we have deemed to include emissions costs.

Table 10. Ongoing costs of measuring Scope 1, Scope 2, and some Scope 3 emissions

Commenter or Source	**Company** (if specified)	**Ongoing costs of measuring Scope 1, Scope 2, and some Scope 3 emissions**
Society for Corporate Governance (June 17, 2022)	Company 9	$200,000 [1]
	Company 10	$83,472 [2]
	Company 11	$75,000 [3]
ERM Survey		$182,985 [4]
Air Products and Chemicals, Inc.		$4,032,000 [5]
Persefoni		$50,000 [6]
S&P Global		$40,000 [7]
South Pole		$23,184 [8]

1. *See* letter from Soc. Corp. Gov. (June 17, 2022). Company 9 discloses Scope 1 and Scope 2 and some Scope 3 GHG emissions but does not specify which categories of Scope 3 emissions are reported. The company "conducts the emissions inventory/data gathering in-house at an estimated cost of at least $200,000 annually." Thus, we assume that $200,000 is the ongoing cost of measuring Scope 1, Scope 2, and some Scope 3 emissions. This estimate may be understated as it is presented as a minimum cost.
2. *See id.* Company 10 discloses Scope 1 and Scope 2 and some Scope 3 (fuel and energy-related activities, business travel, and use of sold products) GHG emissions. Approximately five to seven staff members are involved with the emissions calculations and reporting to various agencies and for verification. The company estimates 188 hours for emissions gathering/annual operating reporting across the company's utility and gas infrastructure business unit and preparing its final verification support. Thus, we assume the 188 burden hours is the ongoing costs of measuring the specified scopes of emissions. (188 hours)*($444/hr) = $83,472.
3. *See id.* Company 11 discloses Scope 1, Scope 2, and some Scope 3 (business travel, commuting, waste, downstream leased assets) GHG emissions. The company estimated its internal time and external resources associated with emissions inventory/data gathering to be about $75,000 annually.
4. *See* ERM survey. The ERM survey indicated that the average spend for GHG analysis and/or disclosures is $237,000 annually. This survey category included all costs related to developing GHG inventories, including analysis and disclosure of Scope 1, Scope 2, and/or Scope 3 emissions. This category also included preparation of GHG data for inclusion in public reporting, any analysis related to setting science-based targets, and other similar efforts to understand GHG emissions. Because this estimate includes targets, we subtract the median ongoing cost of targets ($54,015), as reported in Table 9. $237,000 - $54,015 = $182,985.

5. *See* letter from Air Products and Chemicals, Inc. This company reports Scope 1, Scope 2, and Scope 3 emissions, but does not specify which categories of Scope 3. The company's emissions reporting process requires one full-time consultant and 20 employees working part-time each year from Nov. to Mar. (1 full-time consultant)*(40 hrs/wk)*(20 weeks)*($600/hr) + (20 employees)*(20 hrs/wk)*(20 wks)*($444/hr) = $4,032,000. However, this estimate may be overstated because it includes the cost of third-party limited assurance for GHG emissions.
6. *See supra* note 3023. Persefoni estimates that the cost of assessing Scope 1, Scope 2, and Scope 3 emissions for companies of "high maturity" (i.e., those that are already measuring/tracking Scope 1, Scope 2, and Scope 3 emissions, among other activities) is $50,000, which we assume to reflect ongoing costs. The commenter further estimates that the corresponding cost for companies that do not already measure/track such emissions would be $125,000. If this figure is assumed to represent initial costs, then the estimates reflect a 60% reduction from initial to ongoing costs, which we consider in determining the appropriate percentage reduction in subsequent calculations.
7. *See supra* note 3021. S&P Global estimated that the cost of assessing Scope 1, Scope 2, and Scope 3 emissions for the first time is between $75,000 and $125,000. We take the midpoint of this range ($100,000) and apply the same percent reduction (60%) reflected in the Persefoni meeting memorandum to estimate ongoing costs. $100,000*(1 – 0.6) = $40,000.
8. *See* South Pole Memo. South Pole indicated that conducting a bottom-up assessment of Scope 1, Scope 2, and Scope 3 emissions for the first time can cost between €10,000 and €100,000. We take the midpoint (€55,000), apply the 2022 average exchange rate ($1.0538 USD/€), and apply the same percent reduction (60% reduction) reflected in the Persefoni meeting memorandum to estimate ongoing costs. (€55,000)*(1.0538 USD/€)*(1 – 0.6) = $23,184.

There were other commenters and sources that contained individual cost estimates specific to only Scope 1,[3055] Scopes 1 and 2 combined,[3056] or only Scope 3 emissions measurement,[3057] as opposed to an aggregate cost that combines all three scopes. However, because we determined that all estimates indicated by Column (5) of Table 8 include the aggregate cost of all three scopes of emissions and to remain conservative in our estimation, we opted not to use estimates of individual scopes of emissions for comparability.[3058]

3055 *See* letters from Soc. Corp. Gov. (June 17, 2022); and IDFA.

3056 *See supra* notes 3021, 3023, and 2995; *see also* letter from IDFA.

3057 *See* letters from Williams, Inc.; and Ørsted.

3058 In some cases, commenters' estimates of assessing Scope 1 emissions are greater than other commenters' combined estimates of assessing Scope 1, Scope 2, and some Scope 3 emissions. However, because the resulting adjustment factor will be *subtracted* from Column (3) of Table 8 to obtain compliance costs, we do not include the greater Scope 1 cost estimates in order to remain conservative and to avoid understating final compliance costs. In the following subsection, however, we include these cost estimates when estimating the combined costs of Scopes 1 and 2 emissions in a similar bid to remain conservative.

We have so far obtained the adjustment factors for setting targets and goals ($54,015) and measuring Scope 1, Scope 2, and some Scope 3 emissions ($79,236). We next subtract these amounts, where applicable, from Column (3), from which the result is presented as the adjusted cost in Column (6). The median of the adjusted costs is $183,135. We next extrapolate the initial cost using the assumption of a 44 percent cost reduction[3059] from the first year to subsequent years of these corresponding disclosures. Thus, we estimate that the aggregate compliance costs for governance disclosure; disclosure regarding climate-related risks that have material impacts on strategy, business model, and outlook; and risk management disclosure are $327,000 for the first year and $183,000 for subsequent years (rounded to the nearest $1,000).[3060]

There were additional estimates associated with TCFD disclosure costs that were ultimately not included in this analysis, mainly due to the lack of details needed to obtain a quantitative estimate.[3061] For example, one commenter stated that their "Head of Corporate ESG Strategy and Reporting leads a team of employees that required seven months to gather data and draft disclosures for our 2021 TCFD Report in coordination with numerous subject matter experts across our entire organization."[3062] However, the commenter did not specify how many

---

3059 As noted earlier in this subsection, the FCA rule and BEIS rule reflect a 56 % and 32 % reduction in cost, respectively, from initial year to subsequent years regarding the provisions of interest (*i.e.*, governance disclosure; disclosure regarding climate-related risks that have material impacts on strategy, business model, and outlook; and risk management disclosure). The median, 44 %, is used to estimate the initial cost.

3060 No commenters or sources offered estimates specific to the cost of the disclosure of material expenditures directly related to climate-related activities as part of a registrant's strategy, transition plan and/or targets and goals. Nevertheless, the Commission's estimates (i.e., $327,000 for the first year and $183,000 annually in subsequent years) should reflect this cost based on our application of conservative assumptions and because of the small expected incremental cost given that registrants will likely be tracking the material expenditures under the financial statement disclosure requirements.

3061 *See, e.g.*, letters from Soc. Corp. Gov. (June 11, 2021); Nasdaq; Chamber; and AEPC.

3062 *See* letter from Nasdaq.

staff or FTEs are involved, which precludes us from reliably calculating burden hours and associated costs. Another commenter asserted that the "actual cost for complete alignment to TCFD could be up to $1,000,000 per registrant over several years." Because the commenter did not provide the number of years, however, we are unable to obtain the annual costs.[3063] Other sources provided costs that had general descriptions (*e.g.*, "implementation costs" or "two FTEs… dedicated to climate reporting") that did not explicitly mention "TCFD" disclosures.[3064] We similarly did not include such estimates given that we cannot reliably infer whether these costs are reflective of TCFD disclosures and the specific provisions of interest.

ii. Cost estimates of Scope 1 and 2 emissions disclosures

The final rules require the disclosure of Scope 1 and 2 emissions, if material, by LAFs and AFs, while SRCs and EGCs are exempt.[3065] To inform our assessment of the associated cost, we review comment letters and other sources that contain relevant estimates, presented in Table 11. We note that three of the estimates are specific to the cost of assessing Scope 1 emissions only.[3066] Nevertheless, we include them in Table 11 because (a) these Scope 1 emissions cost estimates are generally higher than other estimates that include both Scope 1 and 2 emissions, and (b) the costs can only increase if the Scope 1 emissions estimates are adjusted to also account for Scope 2 emissions (i.e., they are understated with respect to the cost of both Scope 1 and 2 emissions). Thus, we include the Scope 1 emissions cost estimates to remain conservative in our estimation.

3063 *See* letter from AEPC.

3064 *See, e.g.*, letters from RILA; Nutrien; and Soc. Corp. Gov. (June 11, 2021).

3065 *See* 17 CFR 229.1505(a)(1).

3066 The cost estimates that are specific to Scope 1 only are those from the Society for Corporate Governance (Company 4), Energy Transfer LP, and IDFA ($100,000).

Table 11. Costs of assessing Scope 1 and 2 emissions

Commenter or Source	**Company** (if specified)	**Ongoing cost**
Persefoni		$25,000 [1]
FCA rule		$17,836 [2]
S&P Global		$40,000 [3]
International Dairy Foods Association		$66,600 [4]
International Dairy Foods Association		$100,000 [5]
Energy Transfer LP		$10,162,035 [6]
Society for Corporate Governance (June 17, 2022)	Company 4	$300,000 [7]
Median		**$66,600**